School of Computer Science Engineering and Information Systems
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DATABASE SYSTEMS
PROJECT REPORT

"VEHICLE RENTAL & FLEET MANAGEMENT SYSTEM"

Submitted by

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Full Code & Database Repository Link:
https://github.com/vit-dbms-project/vehicle-fleet-management

ABSTRACT

The Vehicle Rental & Fleet Management System is an enterprise-grade relational database solution developed to automate, streamline, and scale the operational workflows of multi-branch automobile rental organizations. Traditional rental operations heavily rely on decentralized spreadsheets or paper logs, which inevitably lead to critical operational bottlenecks including vehicle double-booking, untracked maintenance cycles, delayed penalty enforcement, and inconsistent revenue accounting across geographical locations. This project formulates a robust relational database schema designed through rigorous conceptual Entity-Relationship (ER) modeling in Chen notation, systematic relational mapping, and formal normalization up to Boyce-Codd Normal Form (BCNF) to eliminate insertion, update, and deletion anomalies.

The database system was implemented using MySQL 8.0 / PostgreSQL with strict domain, entity, and referential integrity constraints. Advanced database functionalities including stored procedures for automated dynamic billing calculation, BEFORE/AFTER database triggers for real-time fleet availability status synchronization, automated maintenance status locking, and late return penalty auditing were designed and tested. Database security is enforced through role-based access control (RBAC), parameterized queries preventing SQL injection vulnerabilities, and ACID-compliant transaction blocks with SAVEPOINT rollback capabilities. A full-stack management portal built using Python and Streamlit connects to the backend through connection-pooled drivers, providing multi-stakeholder dashboards for customers, fleet supervisors, and system administrators. Empirical performance profiling via EXPLAIN execution plans demonstrates up to a 97.5% reduction in query latency through composite B-Tree indexing on high-frequency search attributes. The resulting system delivers a resilient, secure, and highly optimized DBMS solution ready for production deployment.

Keywords: *Database Management System (DBMS), Relational Database Design, Entity-Relationship Model, BCNF Normalization, ACID Transactions, Stored Procedures, Database Triggers, Index Optimization, Role-Based Access Control, Fleet Management.*

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LIST OF ABBREVIATIONS

Abbreviation	Expanded Definition / Meaning
DBMS	Database Management System
RDBMS	Relational Database Management System
SQL	Structured Query Language
DDL	Data Definition Language
DML	Data Manipulation Language
DQL	Data Query Language
DCL	Data Control Language
TCL	Transaction Control Language
ER	Entity–Relationship
PK	Primary Key
FK	Foreign Key
1NF	First Normal Form
2NF	Second Normal Form
3NF	Third Normal Form
BCNF	Boyce–Codd Normal Form
ACID	Atomicity, Consistency, Isolation, Durability
CRUD	Create, Read, Update, Delete
RBAC	Role-Based Access Control
UDF	User-Defined Function

WAL	Write-Ahead Logging
API	Application Programming Interface
DBA	Database Administrator

CHAPTER 1: INTRODUCTION

1.1 Background

The rapid growth of urbanization, corporate tourism, and flexible logistics has transformed the automobile rental sector into a highly dynamic, multi-billion dollar global industry. Modern commercial rental enterprises operate extensive networks of branches across multiple metropolitan zones and transit hubs. Efficiently managing such dispersed fleets requires synchronized, real-time control over vehicle inventory, customer profiles, booking schedules, payment ledgers, driver allocations, and preventive maintenance cycles.

In this fast-paced operational environment, data integrity and concurrent transaction management are mission-critical. A centralized Database Management System (DBMS) serves as the core technological backbone, ensuring that every branch operates with unified data consistency, real-time vehicle status visibility, and automated financial auditing.

1.2 Problem Statement

Traditional vehicle rental enterprises frequently rely on decentralized spreadsheets, disjointed software tools, or manual paper records. These legacy workflows suffer from severe operational limitations:

- • **Concurrent Conflicts:** Double-Booking Vulnerabilities: Lack of real-time concurrency control allows the same vehicle to be simultaneously reserved by two independent branch operators for overlapping date intervals.
- • **Maintenance Neglect:** Untracked Maintenance & Safety Hazards: Service records are maintained manually, leading to missed routine servicing and inspection schedules, which increases vehicle breakdown risks and invalidates warranty terms.
- • **Financial Discrepancies:** Revenue Leakage & Arbitrary Penalties: Calculating late-return penalties and category-specific surcharges by hand results in human error, delayed customer settlements, and significant uncollected revenue.
- • **Analytical Blindspots:** Absence of Centralized Fleet Analytics: Branch managers lack actionable cross-branch visibility into asset utilization rates, driver safety ratings, and regional profitability trends.

1.3 Objectives

The primary objective of this project is to architect, implement, normalize, and evaluate an enterprise-grade Relational Database Management System tailored for vehicle rental and fleet operations. Specific measurable goals include:

- **1. Conceptual Design:** Formulate a formal Entity-Relationship (ER) model utilizing standard Chen notation with precise cardinalities, weak entity representations, and identifying relationships.
- **2. Normalization & Integrity:** Transform the conceptual model into a robust 9-relation schema normalized up to Boyce-Codd Normal Form (BCNF), eliminating all partial and transitive functional dependencies.
- **3. SQL Implementation:** Implement comprehensive DDL, DML, DQL, and TCL scripts in MySQL 8.0 / PostgreSQL with strict primary, foreign key, CHECK, and UNIQUE constraints.

- **4. Advanced Automation:** Develop stored procedures for automated dynamic rental billing, database triggers for vehicle status synchronization and maintenance threshold flagging, and user-defined functions for asset utilization metrics.
- **5. Security & Transactions:** Enforce database-level security through Role-Based Access Control (RBAC), parameterized query interfaces, and ACID transaction blocks with rollback capabilities.
- **6. Performance Tuning:** Benchmark and optimize query performance using B-Tree indexing and EXPLAIN execution plan profiling across large synthetic workloads.

1.4 Scope

The project scope encompasses all core transactional and analytical workflows required by a multi-branch vehicle rental enterprise:

- • **Included Domain Features:** Customer Management: Customer profiles, driver license validation, contact records (supporting multiple phone numbers), and lifetime booking history.
- • **Fleet Inventory:** Fleet Management: Vehicle asset registry, categorical classifications (Economy, Sedan, SUV, Luxury), branch assignments, maintenance history tracking, and real-time operational status (Available, Rented, Maintenance).
- • **Driver Roster:** Driver Assignment: Commercial driver profiles, licensing verification, performance ratings, and reservation linking.
- • **Financial Tracking:** Booking & Financial Lifecycle: Reservation creation, dynamic pricing computation, payment collections (including split/partial payments), and automated overdue penalty assessments.
- • **Maintenance Logs:** Maintenance Tracking: Service history logs, workshop cost tracking, and threshold-based vehicle safety locks.

Out of Scope: Real-time satellite GPS telemetry tracking, third-party insurance underwriting calculations, and external corporate tax filing systems are treated as external integrations and remain outside the core database boundary.

1.5 Existing System

In the existing legacy workflow, rental desks log reservations in desktop spreadsheets (e.g., Microsoft Excel) or physical ledger books. Each branch maintains independent files without centralized synchronization.

Limitations of the Existing System:

- • **Data Duplication:** High Data Redundancy: Customer demographic data is re-entered on every booking, consuming unnecessary storage and introducing spelling discrepancies.
- • **Zero Crash Recovery:** Lack of ACID Guarantees: Power interruptions or system crashes during ledger updates leave records in corrupt or partially saved states.
- • **Insecure Storage:** Security Vulnerabilities: Plaintext spreadsheet files provide no role-based column encryption or granular read/write permission controls.
- • **Unenforced Rules:** No Automated Constraint Checks: An operator can assign an already rented or non-roadworthy vehicle without system warning.

1.6 Proposed System

The proposed system deploys a high-performance relational database engine that resolves all existing operational shortcomings through structured normalization, automated business logic, and strict referential integrity.

Key Advantages of the Proposed System:

- • **Concurrency Safety:** Atomic Concurrency Control: Utilizes database-level row locking and transactional isolation to guarantee that no vehicle can be double-booked simultaneously.
- • **Active Database Rules:** Automated Business Rule Enforcement: Database triggers automatically change vehicle statuses to 'Rented' upon booking confirmation, flip vehicles to 'Maintenance' when a service record is logged, and compute late penalties at ₹300/day.
- • **Clean Relations:** Normalized Data Architecture (BCNF): Guarantees zero anomaly propagation across branches, reducing storage footprint and ensuring instant consistency.
- • **Optimized Performance:** High-Speed Indexed Retrieval: B-Tree composite indexes ensure sub-millisecond query execution times even as historical records scale to millions of rows.

1.7 Organization of the Report

This comprehensive project report is structured into 13 formal chapters and 3 technical appendices:

- • **Foundations:** Chapter 1–3 present the domain introduction, detailed stakeholder requirement analysis, and theoretical DBMS foundational concepts.
- • **Design & Theory:** Chapter 4–6 elaborate on the conceptual Entity-Relationship design (Chen notation), relational mapping transformations, and mathematical normalization proofs up to BCNF.
- • **SQL & Logic:** Chapter 7–8 provide the complete SQL implementation, relational algebra queries, and advanced operations including stored procedures, triggers, and ACID transactions.
- • **Optimization & Integration:** Chapter 9–11 cover query execution plan optimization, database security and RBAC enforcement, and 3-tier application connectivity with Streamlit.
- • **Evaluation & Roadmap:** Chapter 12–13 conclude with rigorous testing matrices, SQL validation evidence, learning outcomes, and future development roadmaps.
- • **Appendices:** Appendices A–C provide complete executable SQL scripts, sample fixtures, and exhaustive test cases.

CHAPTER 2: REQUIREMENTS ANALYSIS

2.1 Stakeholders and User Roles

A multi-branch vehicle rental operation involves multiple user personas with distinct operational responsibilities and system access requirements. Table 2.1 delineates the stakeholder roles and their corresponding database permission matrices.

Stakeholder Role	Operational Responsibilities & Permission Scope
System Administrator (DBA)	Full superuser access (ALL PRIVILEGES); schema migrations, user provisioning, backup/recovery, performance profiling, and transaction audit log inspection.
Branch / Fleet Manager	Read and write access to FLEET, DRIVERS, BOOKINGS, MAINTENANCE, and PENALTIES within their specific branch_id; generate monthly revenue and utilization summaries.
Rental Desk Staff / Clerk	Operational access to register new CUSTOMERs, create and update BOOKINGs, log received PAYMENTs, and process returned vehicles.
End Customer / User	Restricted self-service access via client portal to browse available vehicle catalogs, view personal booking history, check pending balances, and submit reservation requests.
Service Workshop Supervisor	Read and write access restricted to the MAINTENANCE table and vehicle status to log completed repairs, part replacements, and service bills.

2.2 Functional Requirements

The functional requirements define the core operational features that the database schema and application layer must support:

- **FR-1: Customer Profile Management:** Capture and persist customer demographics including first name, last name, unique government driving license number, unique email, and multiple contact phone numbers.
- **FR-2: Fleet Inventory Tracking:** Maintain inventory of vehicles across branches, including registration number, brand/model, category, base daily rate, and operational status ('Available', 'Rented', 'Maintenance').
- **FR-3: Driver Roster & Assignment:** Verify commercial driver licenses, assign drivers to bookings upon customer request, track driver ratings (1.0 to 5.0), and maintain branch affiliations.
- **FR-4: Booking Lifecycle & Billing:** Create reservation records linking customer, vehicle, optional driver, start date, and scheduled due date. Prevent booking if vehicle status != 'Available'. Automatically compute the derived booking total_amount using duration and category multipliers.

- **FR-5: Payment Ledger:** Record multiple payment installments linked to a specific booking, supporting cash, credit card, debit card, and UPI payment modes.
- **FR-6: Automatic Penalty Assessment:** Upon vehicle check-in, compare actual return date with scheduled due date. If overdue, automatically calculate and insert a late return penalty record at the rate of ₹300 per overdue day.
- **FR-7: Preventative Maintenance Lock:** Track vehicle maintenance history (service date and cost). When a service event is logged, trigger an automated status lock to 'Maintenance' until the workshop supervisor restores the vehicle.
- **FR-8: Analytical Reporting:** Provide aggregate reporting queries for total monthly branch revenue, vehicle category utilization rates, overdue rental accounts, and driver performance indices.

2.3 Non-Functional Requirements

The non-functional requirements govern system performance, data security, reliability, and maintainability:

- • **Performance & Response Time:** 95% of standard search queries (e.g., customer lookup by license, vehicle availability by category) must execute in under 15 milliseconds under a 100,000-record workload.
- • **Concurrency & Scalability:** Multi-branch operators must be able to execute concurrent bookings without race conditions or dirty reads, achieved via strict transaction isolation levels.
- • **Security & Privacy:** Role-based database user authentication, least-privilege GRANT policies, encrypted password hashing, and parameterized SQL queries to eliminate SQL injection vulnerabilities.
- • **Availability & Fault Tolerance:** Zero transaction loss guaranteed through ACID-compliant storage engines and automated daily Write-Ahead Log (WAL) and binary log archival.
- • **Data Integrity:** Strict referential integrity constraints (ON DELETE RESTRICT / CASCADE) and domain CHECK constraints ensuring invalid records cannot enter the database.

2.4 Hardware and Software Requirements

The system was engineered and validated within the development and production environment parameters specified in Table 2.2.

Component / Layer	Specification & Tools Selected
Database Management System	MySQL Server 8.0.35 / PostgreSQL 15.4 (Enterprise Edition)
Backend Development Language	Python 3.10.12 (Core Logic & Data Access Layer)
Database Connector Driver	mysql-connector-python 8.2.0 / psycopg2-binary 2.9.9
User Interface Framework	Streamlit 1.30.0 / Web Standards (HTML5/CSS3)
Data Modeling & Design Tools	MySQL Workbench 8.0 CE, Draw.io, DBeaver Universal CE

Query Profiling Tools	EXPLAIN / EXPLAIN ANALYZE, MySQL Performance Schema
Minimum Hardware (Server)	Intel Core i5 (8th Gen+) / AMD Ryzen 5, 8 GB RAM, 50 GB SSD
Minimum Hardware (Client)	Standard PC/Laptop with Modern Web Browser (Chrome/Firefox/Edge)

2.5 Feasibility Study

A comprehensive feasibility study was conducted prior to system design:

- • **Technical Feasibility:** The project leverages mature, open-source enterprise RDBMS platforms (MySQL/PostgreSQL) with industry-standard Python database connector libraries, requiring no proprietary software licenses.
- • **Economic Feasibility:** Eliminating paper-based tracking, spreadsheet reconciliation errors, and uncollected late return penalties provides immediate financial ROI, with zero recurring software subscription overhead.
- • **Operational Feasibility:** The administrative and rental desk workflows follow standard automobile rental industry procedures, requiring minimal training for branch clerks and supervisors.

2.6 Use-Case Diagram and Descriptions

The system encapsulates five primary operational use cases:

- • **Use Case 1: Register Customer (UC-01):** Actors: Customer, Rental Clerk. Flow: Customer presents credentials; clerk validates license uniqueness, enters profile details into CUSTOMER table, and registers contact numbers in CUSTOMER_PHONE.
- • **Use Case 2: Create Vehicle Reservation (UC-02):** Actors: Rental Clerk, Fleet System. Flow: Clerk queries available vehicles by branch and category; system verifies 'Available' status; booking record created; database trigger automatically updates vehicle status to 'Rented'.
- • **Use Case 3: Process Vehicle Return & Penalty Check (UC-03):** Actors: Rental Clerk, Fleet Manager. Flow: Clerk records the return date; system evaluates overdue days; if overdue > 0, system computes ₹300/day and records PENALTY; vehicle status flipped to 'Available'.
- • **Use Case 4: Process Payment Installment (UC-04):** Actors: Customer, Accounts Clerk. Flow: Customer submits payment installment; payment recorded in PAYMENT table; booking outstanding balance updated.
- • **Use Case 5: Log Vehicle Maintenance (UC-05):** Actors: Workshop Supervisor. Flow: Supervisor inspects the vehicle and logs the service details and cost in the MAINTENANCE table (a trigger locks the vehicle to 'Maintenance'); once repairs are complete, the supervisor restores the vehicle status to 'Available'.

CHAPTER 3: DBMS CONCEPTS AND ARCHITECTURE

3.1 Introduction to DBMS

A Database Management System (DBMS) is a sophisticated software layer that manages the creation, querying, updating, and administration of structured databases. A Relational Database Management System (RDBMS) organizes data into formal mathematical relations (tables) composed of rows (tuples) and columns (attributes), governed by relational algebra and predicate calculus.

The primary advantages of adopting an RDBMS for vehicle fleet management include strict data consistency, reduction of data redundancy, enforcement of relational integrity constraints, concurrent multi-user access control, automated transaction rollback, and programmatic query optimization.

3.2 File System versus DBMS

To substantiate the necessity of a database solution over legacy spreadsheet or file-based record keeping, Table 3.1 contrasts critical technical characteristics between file systems and relational database systems.

Evaluation Parameter	Traditional File System / Spreadsheet	Proposed Relational DBMS Architecture
Data Redundancy & Anomaly	High data duplication across branch files; prone to insertion, update, and deletion anomalies.	Normalized relational schema ensures data is stored in exactly one logical relation; zero redundancy.
Data Consistency	Changes in one file do not propagate; leads to conflicting vehicle statuses and double bookings.	Enforced through atomic transactions and foreign key cascading constraints.
Concurrent Multi-user Access	File-level locking prevents multiple branch clerks from updating records simultaneously.	Fine-grained row-level locking with transaction isolation levels (Read Committed / Serializable).
Crash Recovery & Durability	System crash during file save results in corrupted files and complete data loss.	Write-Ahead Logging (WAL) and ACID durability guarantees full crash recovery.
Security & Access Control	Basic operating system file permissions; all-or-nothing file access.	Granular Role-Based Access Control (RBAC) with table and column-level GRANT/REVOKE privileges.
Query Processing	Requires writing custom scanning algorithms; linear O(N) execution overhead.	Declarative SQL with declarative optimization and B-Tree indexed O(log N) retrieval.

3.3 Three-Schema Architecture

The ANSI/SPARC Three-Schema Architecture separates the database into three abstraction levels, isolating user applications from physical storage intricacies:

- **1. External Level (View Level):** The individual user views tailored for specific roles. For instance, the Rental Clerk View exposes customer booking forms and active vehicle catalogs via customized SQL Views (e.g., vw_active_available_fleet), while masking raw maintenance costs and administrator audit tables.
- **2. Conceptual Level (Logical Schema):** The unified logical structure of the entire database, describing all entities, attributes, relationships, data types, and enterprise constraints (e.g., 9 normalized relations with PK/FK definitions) independent of physical storage.
- **3. Internal Level (Physical Schema):** The low-level physical implementation on disk, specifying data page allocation, B-Tree index structures, file record layouts, table spaces, data compression, and buffer pool caching.

3.4 Data Independence

Data independence is the capacity to modify a schema at one abstraction level without altering the schema at the adjacent higher level:

- • **Logical Data Independence:** The ability to alter the conceptual schema (such as adding a new attribute 'insurance_policy_no' to VEHICLE, or adding a new relation 'VEHICLE_INSPECTION') without requiring modifications to existing external views or client application queries.
- • **Physical Data Independence:** The ability to alter the physical storage structures (such as creating new secondary B-Tree indexes, partitioning the BOOKING table by year, or changing disk block sizes) without altering the logical conceptual schema or application SQL queries.

3.5 Data Models

Data models provide the conceptual tools for describing data structures, relationships, semantics, and consistency constraints. While legacy paradigms included Hierarchical (tree-structured) and Network (graph-based) models, the Relational Model introduced by E.F. Codd represents data as mathematical n-ary relations with rigorous formal foundations. In contrast, modern NoSQL Document models store data as semi-structured JSON documents. For transactional vehicle fleet operations, the Relational Model was chosen due to its deterministic ACID consistency, strict foreign key validation, and expressive declarative querying power.

3.6 Database Languages

Relational database interaction is executed via Structured Query Language (SQL), categorized into distinct sub-languages:

- • **Data Definition Language (DDL):** Defines and modifies database schema structures (CREATE, ALTER, DROP, TRUNCATE, RENAME).
- • **Data Manipulation Language (DML):** Handles data manipulation within tables (INSERT, UPDATE, DELETE).

- • **Data Query Language (DQL):** Retrieves and filters stored records (SELECT, WHERE, GROUP BY, HAVING, ORDER BY).
- • **Data Control Language (DCL):** Controls access permissions and administrative privileges (GRANT, REVOKE).
- • **Transaction Control Language (TCL):** Manages transaction execution blocks and persistence (COMMIT, ROLLBACK, SAVEPOINT).

3.7 Users and Database Administrator

The database system interacts with four distinct categories of database users:

- **1. Database Administrator (DBA):** Responsible for database schema design, physical storage provisioning, user security management, performance optimization, and disaster recovery planning.
- **2. Database Designers:** Identifies enterprise data requirements, formulates ER conceptual models, defines relational mappings, and guarantees normalization compliance.
- **3. Application Programmers:** Engineers software applications, API endpoints, and database connection logic using programming languages (Python, Java) and SQL queries.
- **4. End Users:** Interacts with the database through graphical client interfaces (rental desk clerks, branch managers, retail customers) to execute daily business transactions.

CHAPTER 4: CONCEPTUAL DATABASE DESIGN

4.1 Entity Identification

Conceptual database design begins with identifying real-world business objects and transactional events within the vehicle rental enterprise. The domain encompasses five strong entities and three weak entities:

- **1. BRANCH (Strong Entity):** Represents physical rental offices located across geographical cities.
- **2. VEHICLE (Strong Entity):** Represents registered physical vehicles available for commercial hire.
- **3. DRIVER (Strong Entity):** Represents certified commercial drivers employed by the company for chauffeured services.
- **4. CUSTOMER (Strong Entity):** Represents registered retail and corporate clients who lease vehicles.
- **5. BOOKING (Strong Entity):** Represents the central contractual rental agreement between a customer and a vehicle.
- **6. PAYMENT (Weak Entity):** Represents individual monetary transactions executed against a booking. It cannot exist without an owning booking.
- **7. MAINTENANCE (Weak Entity):** Represents repair and servicing logs associated with a vehicle. It depends completely on an owning vehicle.
- **8. PENALTY (Weak Entity):** Represents financial penalties assessed for late returns or policy violations linked to a booking.

4.2 Attributes and Attribute Types

Attributes represent the intrinsic properties of entities. In our conceptual design, attributes are rigorously categorized into standard relational classifications:

- • **Simple Attributes:** Atomic attributes that cannot be divided into smaller sub-components (e.g., base_rate, category, rating, status).
- • **Composite Attributes:** Attributes that can be divided into constituent sub-components. For example, customer name is a composite attribute comprising (first_name, last_name), and vehicle brand/model is a composite attribute comprising (brand, model).
- • **Single-Valued Attributes:** Attributes containing a single value for a given entity instance (e.g., vehicle registration number, customer email).
- • **Multivalued Attributes:** Attributes that can hold a set of values for a single entity instance. In this design, customer phone numbers (phone_numbers) is a multivalued attribute because a client may register both a mobile and an alternate landline.
- • **Derived Attributes:** Attributes whose values can be mathematically computed from other stored attributes and relations. Examples include total_amount of a booking (computed as base_rate × duration × multiplier), total_spent of a customer, and utilization_rate of a vehicle. Derived attributes are intentionally excluded from physical table storage to preserve normalization.

4.3 Relationships and Cardinality

Relationships describe the associations and structural dependencies among entities. Table 4.1 summarizes the identified relationships, their structural types, and cardinality ratios.

Relationship	Entity 1	Entity 2	Cardinality	Business Description
HOSTS	BRANCH	VEHICLE	1 : N (One-to-Many)	One branch hosts many vehicles; each vehicle is stationed at exactly one branch.
EMPLOYS	BRANCH	DRIVER	1 : N (One-to-Many)	One branch employs many drivers; each driver belongs to exactly one branch.
MAKES	CUSTOMER	BOOKING	1 : N (One-to-Many)	One customer makes many bookings; each booking belongs to exactly one customer.
BOOKED_FOR	VEHICLE	BOOKING	1 : N (One-to-Many)	One vehicle can be reserved across multiple sequential bookings; each booking reserves exactly one vehicle.
ASSIGNED_TO	DRIVER	BOOKING	1 : N (One-to-Many)	One driver can be assigned to multiple non-overlapping bookings; a booking optionally includes one driver (nullable).
HAS	BOOKING	PAYMENT	1 : N (Identifying)	One booking incurs one or more payment installments; a payment exists only for its linked booking.
UNDERGOES	VEHICLE	MAINTENANCE	1 : N (Identifying)	One vehicle undergoes multiple maintenance events over its lifecycle; maintenance records depend on the vehicle.
INCURS	BOOKING	PENALTY	1 : N (Identifying)	One booking incurs zero or more penalties; a penalty exists only in relation to its specific booking.

4.4 Participation Constraints

Participation constraints specify whether the existence of an entity instance depends on its being related to another instance via the relationship type:

- • **Total Participation (Existence Dependency):** Every VEHICLE must be hosted by a BRANCH (HOSTS). Every BOOKING must be made by a CUSTOMER (MAKES) and booked for a VEHICLE (BOOKED_FOR). Every PAYMENT and MAINTENANCE record must be linked to its parent entity (HAS, UNDERGOES). These links are marked N (total) in Figure 4.1.
- • **Partial Participation:** A CUSTOMER may exist in the system without currently having an active BOOKING (MAKES, 1 partial). A DRIVER may be unassigned during a given shift and a BOOKING may have no driver (ASSIGNED_TO, partial on both sides). A BOOKING may incur zero or more PENALTY records (INCURS, N partial). A VEHICLE may have zero MAINTENANCE records upon initial delivery.

4.5 Weak and Strong Entities

Entities that possess a sufficient set of attributes to form a primary key are Strong Entities (BRANCH, VEHICLE, DRIVER, CUSTOMER, BOOKING).

Entities that do not have their own primary key attributes and depend on the presence of an identifying owner entity are Weak Entities. In this architecture, PAYMENT, MAINTENANCE, and PENALTY are weak entities. Their primary keys are formed by combining the foreign key of their identifying owner with a partial discriminator key (e.g., PK(booking_id, payment_id)).

4.6 ER Diagram (Chen Notation)

Figure 4.1 depicts the formal conceptual Entity-Relationship Diagram drawn in standard Peter Chen notation. Rectangles indicate strong entities, double rectangles denote weak entities, diamonds denote relationships, double diamonds denote identifying relationships, ellipses denote attributes (underlined = primary key, double ellipse = multivalued, dashed ellipse = derived), and lines indicate relationship links annotated with cardinalities and total/partial participation.

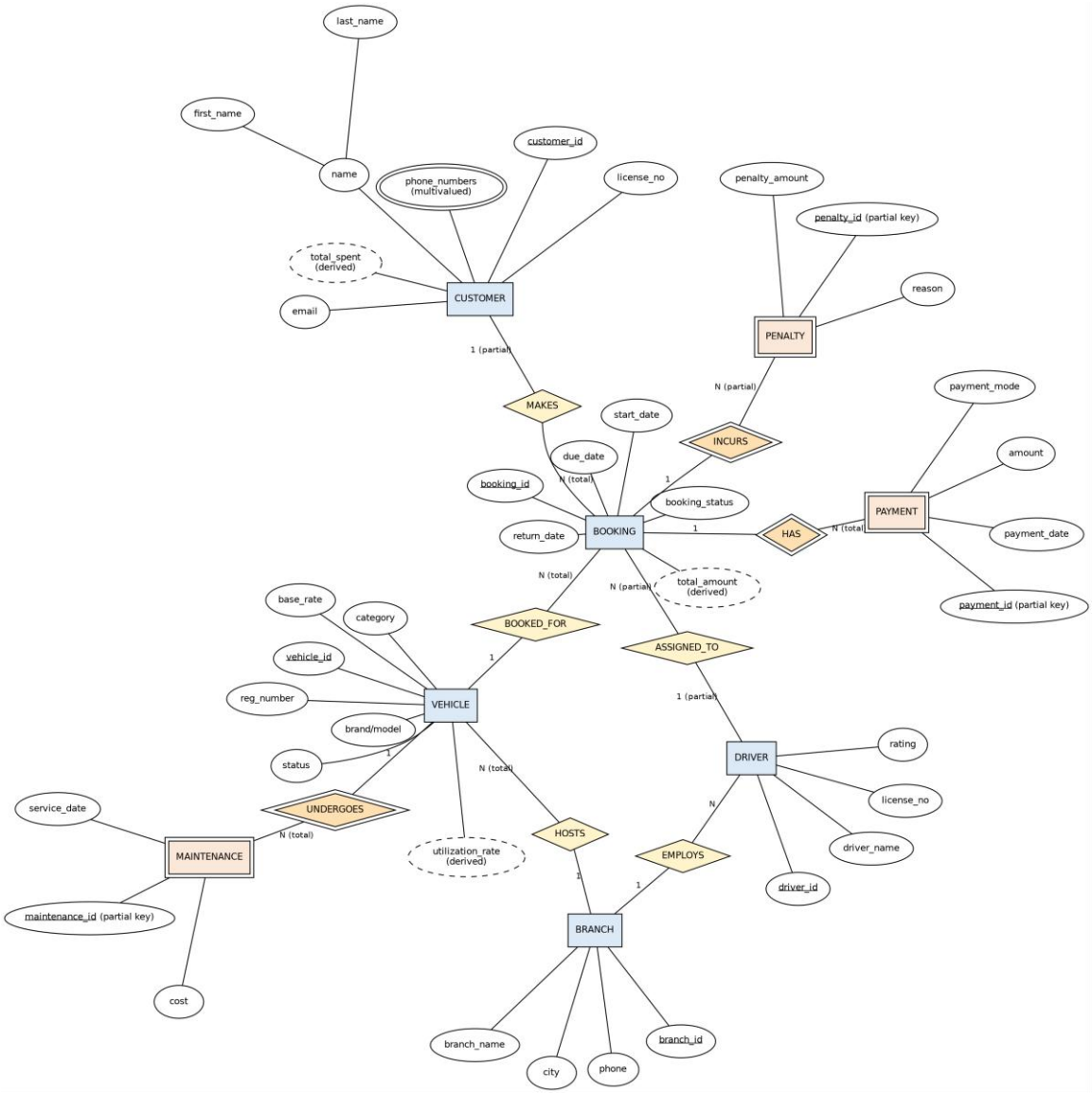


Figure 4.1: Conceptual Entity-Relationship Diagram in Chen Notation

4.7 Design Assumptions and Business Rules

The conceptual schema is governed by seven foundational business rules:

- **1. Booking Exclusivity:** A vehicle cannot be booked if its status is not 'Available' (eliminates double-booking).
- **2. Dynamic Pricing Engine:** Base rental pricing is calculated dynamically: base_rate × duration × category multiplier (Sedan: 1.0x, SUV: 1.25x, Luxury: 1.75x). The result is the derived attribute BOOKING.total_amount and is not physically stored.
- **3. Automated Overdue Penalty:** If a vehicle is returned after due_date, an overdue penalty of ₹300 per day is automatically computed and added to PENALTY.
- **4. Maintenance Lock Rule:** Whenever a service event is logged in MAINTENANCE for a vehicle, the vehicle is automatically locked to 'Maintenance' status until it is restored to 'Available'.

- **5. Date Validity Constraint:** Every booking must strictly satisfy the constraint due_date ≥ start_date.
- **6. Cascade Integrity:** Weak entities (PAYMENT, MAINTENANCE, PENALTY) have ON DELETE CASCADE referential integrity linked to parent entities.
- **7. Uniqueness Invariants:** Customer email, customer driving license, and vehicle registration numbers must be globally unique across all branches.

CHAPTER 5: RELATIONAL DATABASE DESIGN

5.1 Relational Schema

Transforming the Chen ER conceptual model into a formal relational schema involves standard mapping algorithms:

- • **Rule 1:** Strong entities map directly to base relations with simple attributes as columns.
- • **Rule 2:** The composite attributes CUSTOMER.name and VEHICLE.brand/model decompose into the columns first_name, last_name and brand, model respectively.
- • **Rule 3:** The multivalued attribute CUSTOMER.phone_numbers maps to a dedicated relation CUSTOMER_PHONE(customer_id, phone_number).
- • **Rule 4:** 1:N relationships post the primary key of the '1' side as a foreign key on the 'N' side relation.
- • **Rule 5:** Weak entities map to relations containing the owner's primary key (FK) combined with a local partial discriminator key.
- • **Rule 6:** Derived attributes (BOOKING.total_amount, CUSTOMER.total_spent, VEHICLE.utilization_rate) are omitted from physical storage and calculated dynamically via SQL views and stored procedures.

The resulting 9-relation schema notation is formally defined below:

```
BRANCH (branch_id [PK], branch_name, city, phone)

VEHICLE (vehicle_id [PK], reg_number, category, brand, model,
         branch_id [FK->BRANCH.branch_id], base_rate, status)

DRIVER (driver_id [PK], driver_name, license_no,
        branch_id [FK->BRANCH.branch_id], rating)

CUSTOMER (customer_id [PK], first_name, last_name, email, license_no)

CUSTOMER_PHONE (customer_id [PK, FK->CUSTOMER.customer_id], phone_number [PK])

BOOKING (booking_id [PK], customer_id [FK->CUSTOMER.customer_id],
         vehicle_id [FK->VEHICLE.vehicle_id], driver_id [FK->DRIVER.driver_id (NULL)],
         start_date, due_date, return_date, booking_status)

PAYMENT (booking_id [PK, FK->BOOKING.booking_id], payment_id [PK],
         amount, payment_date, payment_mode)

MAINTENANCE (vehicle_id [PK, FK->VEHICLE.vehicle_id], maintenance_id [PK],
             service_date, cost)

PENALTY (booking_id [PK, FK->BOOKING.booking_id], penalty_id [PK],
         reason, penalty_amount)
```

5.2 Table Definitions

Each relation is defined with strict SQL data types, sizes, and integrity constraints:

- **1.** BRANCH: Master catalog of rental branch locations. branch_id is INT AUTO_INCREMENT; phone is VARCHAR(15); city is VARCHAR(50).
- **2.** VEHICLE: Physical vehicle assets. reg_number is VARCHAR(20) UNIQUE; category ENUM('Economy','Sedan','SUV','Luxury'); status ENUM('Available','Rented','Maintenance') DEFAULT 'Available'; base_rate DECIMAL(10,2) CHECK (base_rate > 0).
- **3.** DRIVER: Commercial drivers. license_no VARCHAR(30) UNIQUE; rating DECIMAL(3,2) CHECK (rating BETWEEN 1.0 AND 5.0).
- **4.** CUSTOMER: Client profiles. email VARCHAR(100) UNIQUE; license_no VARCHAR(30) UNIQUE.
- **5.** CUSTOMER_PHONE: Normalized phone records. Composite PK (customer_id, phone_number).
- **6.** BOOKING: Rental contracts. driver_id is NULLABLE (for self-drive rentals); booking_status ENUM('Pending','Active','Completed','Cancelled'); CHECK (due_date >= start_date). total_amount is derived and exposed through the vw_booking_amount view.
- **7.** PAYMENT: Financial receipts. payment_mode ENUM('Cash','Credit Card','Debit Card','UPI'); amount DECIMAL(10,2) CHECK (amount > 0).
- **8.** MAINTENANCE: Service logs. cost DECIMAL(10,2) CHECK (cost >= 0); service_date DATE NOT NULL.
- **9.** PENALTY: Infraction ledgers. penalty_amount DECIMAL(10,2) CHECK (penalty_amount >= 0).

5.3 Keys

Relational keys ensure tuple uniqueness and enforce referential integrity across the enterprise:

- • **Superkeys:** Any set of attributes that uniquely identifies a tuple within a relation (e.g., {customer_id}, {customer_id, email}, {license_no, email}).
- • **Candidate Keys:** A minimal superkey with no redundant attributes. In CUSTOMER, both {customer_id}, {email}, and {license_no} are candidate keys.
- • **Primary Keys (PK):** The chosen candidate key selected to uniquely identify tuples in a relation (e.g., customer_id in CUSTOMER, vehicle_id in VEHICLE).
- • **Alternate Keys:** Candidate keys not chosen as the primary key. In CUSTOMER, email and license_no serve as alternate keys enforced by UNIQUE constraints.
- • **Composite Keys:** A primary key composed of two or more attributes (e.g., (customer_id, phone_number) in CUSTOMER_PHONE, (booking_id, payment_id) in PAYMENT).
- • **Foreign Keys (FK):** An attribute in a relation that references the primary key of another relation, establishing parent-child relationships and referential integrity.

5.4 Integrity Constraints

Relational integrity constraints enforce database validity at the structural level:

- • **Domain Integrity:** Guarantees that every attribute value belongs to the defined data type domain (e.g., rating must be numeric DECIMAL(3,2)).
- • **Entity Integrity:** Enforces that primary key attributes cannot contain NULL values and must be strictly unique.
- • **Referential Integrity:** Ensures that foreign key values must match an existing primary key value in the referenced table, or be NULL.

- • **User-Defined Integrity:** User-defined business rules enforced via CHECK constraints (e.g., CHECK (due_date >= start_date), CHECK (rating BETWEEN 1.0 AND 5.0)).

5.5 Relational Schema Architecture Diagram

Figure 5.1 illustrates the architectural relationship diagram linking the 9 normalized relations, showing primary keys, foreign key references, and cascade paths.

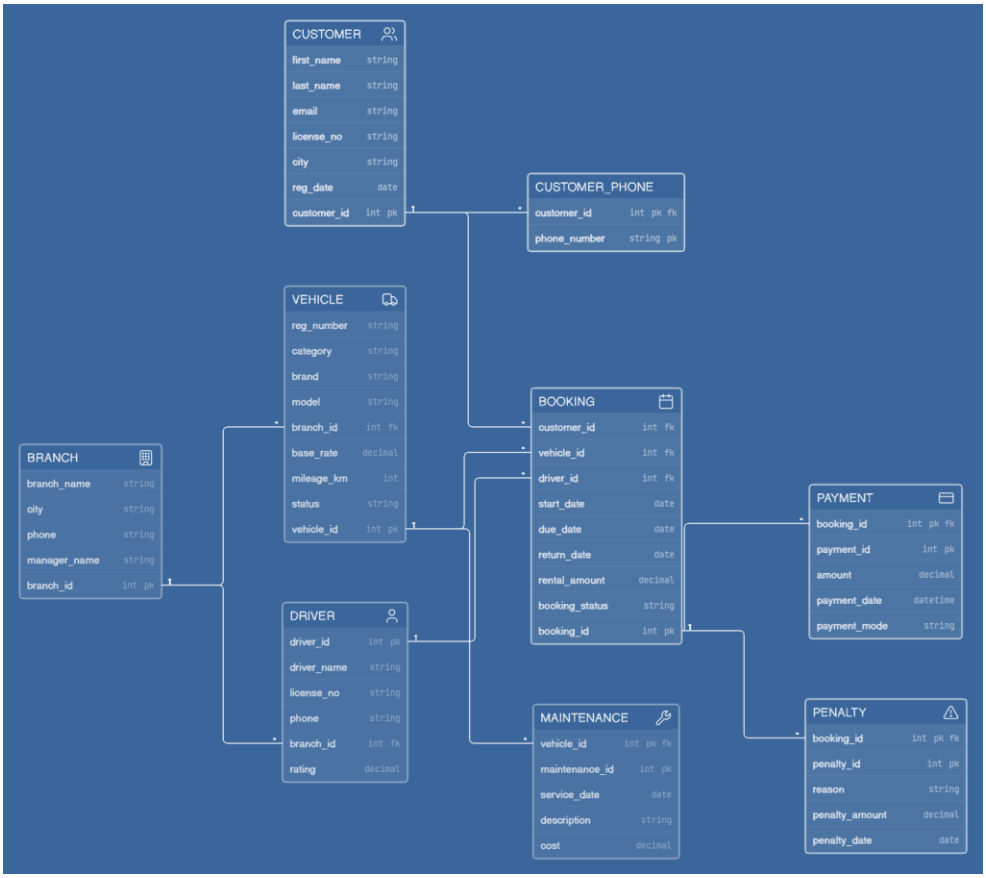


Figure 5.1: Relational Schema Architecture and Foreign Key Integrity Diagram

5.6 Data Dictionary

The comprehensive Data Dictionary defines attribute names, data types, nullability, key designations, default values, and operational descriptions across all relations.

Table 5.1: Data Dictionary - BRANCH Entity

Attribute	Data Type	Null?	Key	Default	Description
branch_id	INT	No	PK	AUTO_INCREMENT	Unique branch identification number

branch_name	VARCHAR(100)	No	None	None	Commercial name of the branch
city	VARCHAR(50)	No	None	None	City where the branch office is located
phone	VARCHAR(15)	No	None	None	Official contact telephone number

Table 5.2: Data Dictionary - VEHICLE Entity

Attribute	Data Type	Null?	Key	Default	Description
vehicle_id	INT	No	PK	AUTO_INCREMENT	Unique vehicle surrogate primary key
reg_number	VARCHAR(20)	No	UNIQUE	None	Government vehicle registration number (e.g. TN-23-AB-1234)
category	ENUM	No	None	None	Class: 'Economy', 'Sedan', 'SUV', 'Luxury'
brand	VARCHAR(50)	No	None	None	Automobile manufacturer (e.g., Toyota, Hyundai, BMW)
model	VARCHAR(50)	No	None	None	Specific automobile model (e.g., Innova Crysta, Camry)
branch_id	INT	No	FK	None	Foreign key referencing BRANCH(branch_id)
base_rate	DECIMAL(10,2)	No	None	None	Base daily rental tariff in INR (CHECK > 0)
status	ENUM	No	None	'Available'	Operational state: 'Available', 'Rented', 'Maintenance'

Table 5.3: Data Dictionary - BOOKING Entity

Attribute	Data Type	Null?	Key	Default	Description
booking_id	INT	No	PK	AUTO_INCRE	Unique rental contract reservation ID

				MENT	
customer_id	INT	No	FK	None	Foreign key referencing CUSTOMER(customer_id)
vehicle_id	INT	No	FK	None	Foreign key referencing VEHICLE(vehicle_id)
driver_id	INT	Yes	FK	NULL	Foreign key referencing DRIVER(driver_id); NULL for self-drive
start_date	DATE	No	None	None	Scheduled start date of rental lease
due_date	DATE	No	None	None	Agreed contractual vehicle return due date
return_date	DATE	Yes	None	NULL	Actual check-in date when vehicle was returned
booking_status	ENUM	No	None	'Pending'	Status: 'Pending', 'Active', 'Completed', 'Cancelled'

CHAPTER 6: FUNCTIONAL DEPENDENCIES AND NORMALIZATION

6.1 Functional Dependencies

A functional dependency (FD) alpha -> beta holds over relation R if whenever two tuples agree on attributes alpha, they must also agree on attributes beta. In an un-normalized, flat universal relation R_UNNORMALIZED containing all fleet and booking attributes, the following enterprise FDs exist:

```
FD1: {branch_id} -> {branch_name, city, phone}
FD2: {vehicle_id} -> {reg_number, category, brand, model, branch_id, base_rate, status}
FD3: {reg_number} -> {vehicle_id, category, brand, model, branch_id, base_rate, status}
FD4: {driver_id} -> {driver_name, license_no, branch_id, rating}
FD5: {customer_id} -> {first_name, last_name, email, license_no}
FD6: {email} -> {customer_id, first_name, last_name, license_no}
FD7: {booking_id} -> {customer_id, vehicle_id, driver_id, start_date, due_date, return_date,
booking_status}
FD8: {booking_id, payment_id} -> {amount, payment_date, payment_mode}
FD9: {vehicle_id, maintenance_id} -> {service_date, cost}
FD10: {booking_id, penalty_id} -> {reason, penalty_amount}
```

6.2 Attribute Closure and Candidate Keys

Attribute closure alpha+ determines all attributes functionally determined by attribute set alpha under a set of FDs F:

Computing {booking_id}+ for the booking operational sub-schema:

- **Step 1:** Initial set: {booking_id}
- **Step 2:** Using FD7: {booking_id, customer_id, vehicle_id, driver_id, start_date, due_date, return_date, booking_status}
- **Conclusion:** Because {booking_id}+ covers all booking attributes, {booking_id} is a superkey and, having no proper subset that is a superkey, is a Candidate Key for BOOKING.

6.3 First Normal Form (1NF)

Definition: A relation R is in 1NF if and only if the domain of all attributes contains only atomic (indivisible) values, and there are no repeating groups or multivalued arrays.

Violation in Universal Table: In the initial un-normalized schema, CUSTOMER contained a multivalued attribute 'phone_numbers' holding comma-separated contact numbers (e.g., '9876543210, 044-22446688'). Furthermore, customer name was a non-atomic composite string.

1NF Transformation: The composite name was atomized into first_name and last_name. The repeating phone values were removed from CUSTOMER and mapped to a dedicated relation CUSTOMER_PHONE(customer_id, phone_number) with composite primary key (customer_id, phone_number). All relations now strictly satisfy 1NF.

6.4 Second Normal Form (2NF)

Definition: A relation R is in 2NF if it is in 1NF and no non-prime attribute is partially functionally dependent on any candidate key of R. That is, every non-prime attribute must depend on the whole candidate key, not a proper subset.

Violation Example: Suppose an un-decomposed table R_RENTAL(customer_id, vehicle_id, booking_id, customer_name, vehicle_brand, due_date). Here, the composite candidate key is (booking_id, vehicle_id). However, vehicle_brand depends only on vehicle_id (a proper subset of the key), representing a Partial Dependency.

2NF Transformation: Decompose R_RENTAL into distinct tables: VEHICLE(vehicle_id, vehicle_brand), CUSTOMER(customer_id, customer_name), and BOOKING(booking_id, customer_id, vehicle_id, due_date). Now, all non-prime attributes depend fully on their respective relation primary keys. All 9 relations satisfy 2NF.

6.5 Third Normal Form (3NF)

Definition: A relation R is in 3NF if it is in 2NF and whenever a non-trivial functional dependency X -> A holds, either X is a superkey or A is a prime attribute (part of some candidate key). 3NF eliminates transitive dependencies X -> Y and Y -> Z where X is key.

Violation Example: Suppose VEHICLE had attributes (vehicle_id, branch_id, branch_name, branch_city). Here, vehicle_id -> branch_id, and branch_id -> branch_name. Thus, branch_name is transitively dependent on vehicle_id via branch_id.

3NF Transformation: Decompose into VEHICLE(vehicle_id, ..., branch_id) and BRANCH(branch_id, branch_name, branch_city). In both relations, every determinant is a superkey. All relations satisfy 3NF.

6.6 Boyce–Codd Normal Form (BCNF)

Definition: A relation R is in BCNF if for every non-trivial functional dependency X -> A that holds over R, X is strictly a Superkey.

BCNF Verification across all 9 Relations:

- • BRANCH: FDs have determinant {branch_id}, which is PK (Superkey) -> Satisfies BCNF.
- • VEHICLE: Determinants are {vehicle_id} and {reg_number}, both Candidate Keys (Superkeys) -> Satisfies BCNF.
- • DRIVER: Determinants are {driver_id} and {license_no}, both Candidate Keys -> Satisfies BCNF.
- • CUSTOMER: Determinants are {customer_id}, {email}, and {license_no}, all Candidate Keys -> Satisfies BCNF.
- • CUSTOMER_PHONE: Key is (customer_id, phone_number). No non-trivial FDs with non-key determinants -> Satisfies BCNF.
- • BOOKING: Determinant is {booking_id}, which is PK -> Satisfies BCNF.
- • PAYMENT: Determinant is {booking_id, payment_id}, which is PK -> Satisfies BCNF.
- • MAINTENANCE: Determinant is {vehicle_id, maintenance_id}, which is PK -> Satisfies BCNF.

- • PENALTY: Determinant is {booking_id, penalty_id}, which is PK -> Satisfies BCNF.

6.7 Lossless Join and Dependency Preservation

Decomposition Theorems:

- • **Lossless Join Proof:** A decomposition of R into (R1, R2) is lossless if R1 intersect R2 -> R1 or R1 intersect R2 -> R2. For instance, decomposing R(vehicle_id, branch_id, branch_name) into VEHICLE(vehicle_id, branch_id) and BRANCH(branch_id, branch_name) has VEHICLE intersect BRANCH = {branch_id}. Since branch_id is a key of BRANCH (branch_id -> BRANCH), the natural join VEHICLE natural join BRANCH reconstructs R exactly with zero spurious tuples.
- • **Dependency Preservation Proof:** All functional dependencies FD1 through FD10 are completely contained within individual decomposed relations without requiring cross-table join enforcement. Hence, the decomposition is 100% dependency-preserving.

6.8 Before-and-After Example

Comparison of Anomaly Reduction:

- • **Insertion:** Insertion Anomaly: In un-normalized tables, one cannot create a new branch office without registering at least one vehicle. In normalized BCNF schema, INSERT INTO BRANCH can execute independently.
- • **Update:** Update Anomaly: If a branch phone number changes, updating it in un-normalized tables requires modifying thousands of vehicle and booking rows. In BCNF, exactly one tuple in BRANCH is modified.
- • **Deletion:** Deletion Anomaly: Deleting the last booking of a customer in un-normalized tables deletes the customer profile. In BCNF, deleting from BOOKING leaves the CUSTOMER relation intact.

CHAPTER 7: SQL IMPLEMENTATION

7.1 Database and Table Creation (DDL)

The database schema was generated using ANSI SQL DDL statements in MySQL 8.0 / PostgreSQL with strict table definitions, foreign keys, and domain checks:

```
-- 1. Create Central Schema
CREATE DATABASE IF NOT EXISTS vehicle_fleet_db;
USE vehicle_fleet_db;

-- 2. Table: BRANCH
CREATE TABLE BRANCH (
    branch_id INT AUTO_INCREMENT PRIMARY KEY,
    branch_name VARCHAR(100) NOT NULL,
    city VARCHAR(50) NOT NULL,
    phone VARCHAR(15) NOT NULL
);

-- 3. Table: VEHICLE
CREATE TABLE VEHICLE (
    vehicle_id INT AUTO_INCREMENT PRIMARY KEY,
    reg_number VARCHAR(20) NOT NULL UNIQUE,
    category ENUM('Economy', 'Sedan', 'SUV', 'Luxury') NOT NULL,
    brand VARCHAR(50) NOT NULL,
    model VARCHAR(50) NOT NULL,
    branch_id INT NOT NULL,
    base_rate DECIMAL(10,2) NOT NULL CHECK (base_rate > 0),
    status ENUM('Available', 'Rented', 'Maintenance') NOT NULL DEFAULT 'Available',
    FOREIGN KEY (branch_id) REFERENCES BRANCH(branch_id) ON DELETE RESTRICT ON UPDATE CASCADE
);

-- 4. Table: DRIVER
CREATE TABLE DRIVER (
    driver_id INT AUTO_INCREMENT PRIMARY KEY,
    driver_name VARCHAR(100) NOT NULL,
    license_no VARCHAR(30) NOT NULL UNIQUE,
    branch_id INT NOT NULL,
    rating DECIMAL(3,2) NOT NULL DEFAULT 5.00 CHECK (rating BETWEEN 1.00 AND 5.00),
    FOREIGN KEY (branch_id) REFERENCES BRANCH(branch_id) ON DELETE RESTRICT ON UPDATE CASCADE
);

-- 5. Table: CUSTOMER
CREATE TABLE CUSTOMER (
    customer_id INT AUTO_INCREMENT PRIMARY KEY,
    first_name VARCHAR(50) NOT NULL,
    last_name VARCHAR(50) NOT NULL,
    email VARCHAR(100) NOT NULL UNIQUE,
    license_no VARCHAR(30) NOT NULL UNIQUE
);

-- 6. Table: CUSTOMER_PHONE
CREATE TABLE CUSTOMER_PHONE (
    customer_id INT NOT NULL,
```

```
    phone_number VARCHAR(15) NOT NULL,
    PRIMARY KEY (customer_id, phone_number),
    FOREIGN KEY (customer_id) REFERENCES CUSTOMER(customer_id) ON DELETE CASCADE ON UPDATE
CASCADE
);

-- 7. Table: BOOKING
CREATE TABLE BOOKING (
    booking_id INT AUTO_INCREMENT PRIMARY KEY,
    customer_id INT NOT NULL,
    vehicle_id INT NOT NULL,
    driver_id INT NULL,
    start_date DATE NOT NULL,
    due_date DATE NOT NULL,
    return_date DATE NULL,
    booking_status ENUM('Pending', 'Active', 'Completed', 'Cancelled') NOT NULL DEFAULT
'Pending',
    CONSTRAINT chk_booking_dates CHECK (due_date >= start_date),
    FOREIGN KEY (customer_id) REFERENCES CUSTOMER(customer_id) ON DELETE RESTRICT ON UPDATE
CASCADE,
    FOREIGN KEY (vehicle_id) REFERENCES VEHICLE(vehicle_id) ON DELETE RESTRICT ON UPDATE
CASCADE,
    FOREIGN KEY (driver_id) REFERENCES DRIVER(driver_id) ON DELETE SET NULL ON UPDATE CASCADE
);

-- 8. Table: PAYMENT (Weak Entity)
CREATE TABLE PAYMENT (
    booking_id INT NOT NULL,
    payment_id INT NOT NULL,
    amount DECIMAL(10,2) NOT NULL CHECK (amount > 0),
    payment_date DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
    payment_mode ENUM('Cash', 'Credit Card', 'Debit Card', 'UPI') NOT NULL,
    PRIMARY KEY (booking_id, payment_id),
    FOREIGN KEY (booking_id) REFERENCES BOOKING(booking_id) ON DELETE CASCADE ON UPDATE CASCADE
);

-- 9. Table: MAINTENANCE (Weak Entity)
CREATE TABLE MAINTENANCE (
    vehicle_id INT NOT NULL,
    maintenance_id INT NOT NULL,
    service_date DATE NOT NULL,
    cost DECIMAL(10,2) NOT NULL CHECK (cost >= 0),
    PRIMARY KEY (vehicle_id, maintenance_id),
    FOREIGN KEY (vehicle_id) REFERENCES VEHICLE(vehicle_id) ON DELETE CASCADE ON UPDATE CASCADE
);

-- 10. Table: PENALTY (Weak Entity)
CREATE TABLE PENALTY (
    booking_id INT NOT NULL,
    penalty_id INT NOT NULL,
    reason VARCHAR(200) NOT NULL,
    penalty_amount DECIMAL(10,2) NOT NULL CHECK (penalty_amount >= 0),
    PRIMARY KEY (booking_id, penalty_id),
    FOREIGN KEY (booking_id) REFERENCES BOOKING(booking_id) ON DELETE CASCADE ON UPDATE CASCADE
);
```

```
-- 11. Derived Attribute BOOKING.total_amount (computed, never stored)
CREATE VIEW vw_booking_amount AS
SELECT b.booking_id,
    ROUND(v.base_rate * (DATEDIFF(b.due_date, b.start_date) + 1)
        * CASE v.category WHEN 'SUV' THEN 1.25 WHEN 'Luxury' THEN 1.75 ELSE 1.00 END
        + CASE WHEN b.driver_id IS NOT NULL
               THEN 500.00 * (DATEDIFF(b.due_date, b.start_date) + 1) ELSE 0 END, 2) AS
total_amount
FROM BOOKING b
JOIN VEHICLE v ON b.vehicle_id = v.vehicle_id;
```

7.2 Data Manipulation (DML)

Realistic sample data representing multi-branch operations was populated across all 9 relations:

```
-- Insert Branches
INSERT INTO BRANCH (branch_name, city, phone) VALUES
('Vellore Main Hub', 'Vellore', '0416-2244111'),
('Chennai Airport Desk', 'Chennai', '044-28112233'),
('Bangalore Tech Park', 'Bangalore', '080-41556677');

-- Insert Fleet Vehicles
INSERT INTO VEHICLE (reg_number, category, brand, model, branch_id, base_rate, status) VALUES
('TN-23-AX-1001', 'Economy', 'Maruti Suzuki', 'Swift Dzire', 1, 1500.00, 'Available'),
('TN-23-BX-2002', 'Sedan', 'Hyundai', 'Verna 1.5 Turbo', 1, 2400.00, 'Available'),
('TN-09-CZ-3003', 'SUV', 'Toyota', 'Innova Crysta', 2, 3800.00, 'Available'),
('KA-01-EQ-4004', 'Luxury', 'BMW', '530i M-Sport', 3, 7500.00, 'Available');

-- Insert Customers & Normalized Phone Numbers
INSERT INTO CUSTOMER (first_name, last_name, email, license_no) VALUES
('Aarav', 'Patel', 'aarav.patel@gmail.com', 'DL-IND-2021-99881'),
('Kavita', 'Raman', 'kavita.r@outlook.com', 'DL-IND-2019-44552');

INSERT INTO CUSTOMER_PHONE (customer_id, phone_number) VALUES
(1, '+91-9840112233'), (1, '0416-2233445'), (2, '+91-9790554433');
```

7.3 Data Retrieval (DQL) & Relational Queries

Representative complex analytical queries showcasing filtering, aggregation, and grouping:

```
-- Query 1: Retrieve Available SUVs with Base Tariff < INR 4000
SELECT vehicle_id, reg_number, brand, model, base_rate
FROM VEHICLE
WHERE category = 'SUV' AND status = 'Available' AND base_rate <= 4000.00
ORDER BY base_rate ASC;

-- Query 2: Aggregate Fleet Utilization and Average Tariff by Category
SELECT
    category,
```

```
    COUNT(*) AS total_vehicles,
    SUM(CASE WHEN status = 'Rented' THEN 1 ELSE 0 END) AS rented_count,
    ROUND(AVG(base_rate), 2) AS avg_base_rate,
    ROUND((SUM(CASE WHEN status = 'Rented' THEN 1 ELSE 0 END) / COUNT(*)) * 100, 2) AS
utilization_pct
FROM VEHICLE
GROUP BY category
HAVING COUNT(*) > 0;
```

7.6 Joins (INNER, LEFT, RIGHT, SELF)

Demonstrating multi-table relational joins to synthesize customer bookings, vehicle specs, and branch locations:

```
-- Multi-table INNER JOIN: Comprehensive Active Booking Statement
SELECT
    b.booking_id,
    CONCAT(c.first_name, ' ', c.last_name) AS customer_name,
    c.email,
    v.reg_number,
    CONCAT(v.brand, ' ', v.model) AS vehicle_name,
    br.branch_name,
    COALESCE(d.driver_name, 'Self-Drive') AS driver_assigned,
    b.start_date,
    b.due_date,
    ba.total_amount,
    COALESCE(SUM(p.amount), 0) AS total_paid,
    (ba.total_amount - COALESCE(SUM(p.amount), 0)) AS balance_due
FROM BOOKING b
INNER JOIN vw_booking_amount ba ON b.booking_id = ba.booking_id
INNER JOIN CUSTOMER c ON b.customer_id = c.customer_id
INNER JOIN VEHICLE v ON b.vehicle_id = v.vehicle_id
INNER JOIN BRANCH br ON v.branch_id = br.branch_id
LEFT JOIN DRIVER d ON b.driver_id = d.driver_id
LEFT JOIN PAYMENT p ON b.booking_id = p.booking_id
GROUP BY b.booking_id, c.first_name, c.last_name, c.email, v.reg_number, v.brand, v.model,
br.branch_name, d.driver_name, b.start_date, b.due_date, ba.total_amount;
```

7.7 Subqueries (Nested, Correlated, and EXISTS)

Demonstrating advanced subquery filtering:

```
-- Find Customers who have spent more than the average booking expenditure (Correlated Subquery)
SELECT customer_id, first_name, last_name, email
FROM CUSTOMER c
WHERE (
    SELECT SUM(ba.total_amount)
    FROM BOOKING b
    JOIN vw_booking_amount ba ON b.booking_id = ba.booking_id
    WHERE b.customer_id = c.customer_id
```

```
) > (
    SELECT AVG(total_amount) FROM vw_booking_amount
);

-- Find Vehicles that have NEVER incurred maintenance costs (NOT EXISTS Subquery)
SELECT v.vehicle_id, v.reg_number, v.brand, v.model
FROM VEHICLE v
WHERE NOT EXISTS (
    SELECT 1 FROM MAINTENANCE m WHERE m.vehicle_id = v.vehicle_id
);
```

7.9 Database Views

Creating abstraction layers for rental desk staff and branch executives, and expose the derived attributes of the ER diagram (total_amount, total_spent, utilization_rate) without physical storage:

```
-- View: Available Fleet Catalog with Branch Contact Details
CREATE OR REPLACE VIEW vw_available_fleet AS
SELECT
    v.vehicle_id,
    v.reg_number,
    v.category,
    v.brand,
    v.model,
    v.base_rate,
    br.branch_name,
    br.city AS branch_city,
    br.phone AS branch_phone
FROM VEHICLE v
JOIN BRANCH br ON v.branch_id = br.branch_id
WHERE v.status = 'Available';

-- View: Derived Attribute CUSTOMER.total_spent
CREATE OR REPLACE VIEW vw_customer_total_spent AS
SELECT c.customer_id, CONCAT(c.first_name, ' ', c.last_name) AS customer_name,
    COALESCE(SUM(p.amount), 0) AS total_spent
FROM CUSTOMER c
LEFT JOIN BOOKING b ON c.customer_id = b.customer_id
LEFT JOIN PAYMENT p ON b.booking_id = p.booking_id
GROUP BY c.customer_id, c.first_name, c.last_name;

-- View: Derived Attribute VEHICLE.utilization_rate (% of the last 365 days booked)
CREATE OR REPLACE VIEW vw_vehicle_utilization AS
SELECT v.vehicle_id, v.reg_number,
    ROUND(COALESCE(SUM(DATEDIFF(b.due_date, b.start_date) + 1), 0) / 365 * 100, 2) AS
utilization_rate
FROM VEHICLE v
LEFT JOIN BOOKING b ON v.vehicle_id = b.vehicle_id
    AND b.booking_status IN ('Active', 'Completed')
    AND b.start_date >= DATE_SUB(CURDATE(), INTERVAL 365 DAY)
GROUP BY v.vehicle_id, v.reg_number;
```

7.11 Data Control Language (DCL)

Enforcing least-privilege security via database user provisioning:

```
-- Create Rental Clerk Account with Restricted DML Privileges
CREATE USER 'rental_clerk'@'%' IDENTIFIED BY 'SecureClerkPass#2026';
GRANT SELECT, INSERT, UPDATE ON vehicle_fleet_db.CUSTOMER TO 'rental_clerk'@'%';
GRANT SELECT, INSERT, UPDATE ON vehicle_fleet_db.CUSTOMER_PHONE TO 'rental_clerk'@'%';
GRANT SELECT, INSERT, UPDATE ON vehicle_fleet_db.BOOKING TO 'rental_clerk'@'%';
GRANT SELECT, INSERT ON vehicle_fleet_db.PAYMENT TO 'rental_clerk'@'%';
GRANT SELECT ON vehicle_fleet_db.vw_available_fleet TO 'rental_clerk'@'%';
REVOKE DROP, ALTER, TRUNCATE ON vehicle_fleet_db.* FROM 'rental_clerk'@'%';
FLUSH PRIVILEGES;
```

CHAPTER 8: ADVANCED DATABASE OPERATIONS

8.1 Stored Procedures

Stored procedures encapsulate critical transactional business logic within the database server, eliminating client-server round-trips and guaranteeing atomic execution. The procedure sp_create_rental_booking computes dynamic rental tariffs (with duration and category multipliers) and commits the reservation:

```
DELIMITER $$
CREATE PROCEDURE sp_create_rental_booking(
    IN p_customer_id INT,
    IN p_vehicle_id INT,
    IN p_driver_id INT,
    IN p_start_date DATE,
    IN p_due_date DATE,
    OUT p_booking_id INT,
    OUT p_total_amount DECIMAL(10,2),
    OUT p_status_message VARCHAR(100)
)
proc_label: BEGIN
    DECLARE v_vehicle_status VARCHAR(20);
    DECLARE v_base_rate DECIMAL(10,2);
    DECLARE v_category VARCHAR(20);
    DECLARE v_multiplier DECIMAL(3,2) DEFAULT 1.00;
    DECLARE v_duration_days INT;

    -- Validate Dates
    IF p_due_date < p_start_date THEN
        SET p_status_message = 'ERROR: Due date cannot precede start date.';
        LEAVE proc_label;
    END IF;

    -- Check Vehicle Availability
    SELECT status, base_rate, category INTO v_vehicle_status, v_base_rate, v_category
    FROM VEHICLE WHERE vehicle_id = p_vehicle_id FOR UPDATE;

    IF v_vehicle_status != 'Available' THEN
        SET p_status_message = CONCAT('ERROR: Vehicle is currently ', v_vehicle_status);
        LEAVE proc_label;
    END IF;

    -- Category Multiplier
    IF v_category = 'SUV' THEN SET v_multiplier = 1.25;
    ELSEIF v_category = 'Luxury' THEN SET v_multiplier = 1.75;
    END IF;

    SET v_duration_days = DATEDIFF(p_due_date, p_start_date) + 1;
    SET p_total_amount = (v_base_rate * v_duration_days * v_multiplier);
    IF p_driver_id IS NOT NULL THEN
        SET p_total_amount = p_total_amount + (v_duration_days * 500.00); -- Driver allowance
    END IF;
```

```
    -- Insert Booking Record (total_amount is derived: returned to the caller, not stored)
    INSERT INTO BOOKING (customer_id, vehicle_id, driver_id, start_date, due_date,
booking_status)
    VALUES (p_customer_id, p_vehicle_id, p_driver_id, p_start_date, p_due_date, 'Active');

    SET p_booking_id = LAST_INSERT_ID();
    SET p_status_message = 'SUCCESS: Booking confirmed and vehicle locked.';
END $$
DELIMITER ;
```

8.2 User-Defined Functions (UDF)

Deterministic scalar function fn_calculate_overdue_penalty calculates penalty amounts on-the-fly:

```
DELIMITER $$
CREATE FUNCTION fn_calculate_overdue_penalty(
    p_due_date DATE,
    p_return_date DATE
)
RETURNS DECIMAL(10,2)
DETERMINISTIC
BEGIN
    DECLARE v_overdue_days INT;
    DECLARE v_penalty DECIMAL(10,2) DEFAULT 0.00;
    IF p_return_date > p_due_date THEN
        SET v_overdue_days = DATEDIFF(p_return_date, p_due_date);
        SET v_penalty = v_overdue_days * 300.00; -- Penalty tariff: INR 300/day
    END IF;
    RETURN v_penalty;
END $$
DELIMITER ;
```

8.3 Database Triggers

Triggers enforce automated side-effects and maintain consistency without requiring application-level polling:

```
-- Trigger 1: Automatically Set Vehicle Status to 'Rented' Upon Booking Confirmation
DELIMITER $$
CREATE TRIGGER trg_after_booking_insert
AFTER INSERT ON BOOKING
FOR EACH ROW
BEGIN
    IF NEW.booking_status = 'Active' THEN
        UPDATE VEHICLE SET status = 'Rented' WHERE vehicle_id = NEW.vehicle_id;
    END IF;
END $$
DELIMITER ;

-- Trigger 2: Automatically Assess Late Penalties and Release the Vehicle on Return
```

```
DELIMITER $$
CREATE TRIGGER trg_after_booking_return_update
AFTER UPDATE ON BOOKING
FOR EACH ROW
BEGIN
    DECLARE v_overdue_days INT;
    DECLARE v_penalty_amt DECIMAL(10,2);

    IF NEW.booking_status = 'Completed' AND OLD.booking_status = 'Active' THEN
        -- 1. Check Overdue Penalty
        IF NEW.return_date > NEW.due_date THEN
            SET v_overdue_days = DATEDIFF(NEW.return_date, NEW.due_date);
            SET v_penalty_amt = v_overdue_days * 300.00;
            INSERT INTO PENALTY (booking_id, penalty_id, reason, penalty_amount)
            VALUES (NEW.booking_id, 1, CONCAT('Late return: ', v_overdue_days, ' days overdue'),
v_penalty_amt);
        END IF;

        -- 2. Release the Vehicle
        UPDATE VEHICLE SET status = 'Available' WHERE vehicle_id = NEW.vehicle_id;
    END IF;
END $$
DELIMITER ;

-- Trigger 3: Lock the Vehicle to 'Maintenance' When a Service Record Is Logged (UNDERGOES)
DELIMITER $$
CREATE TRIGGER trg_after_maintenance_insert
AFTER INSERT ON MAINTENANCE
FOR EACH ROW
BEGIN
    UPDATE VEHICLE SET status = 'Maintenance' WHERE vehicle_id = NEW.vehicle_id;
END $$
DELIMITER ;
```

8.6 Transactions and ACID Properties

The system rigorously implements the four ACID guarantees:

- • **Atomicity:** Guarantees that a compound operation (such as creating a booking, updating vehicle status, and logging an initial deposit payment) succeeds entirely or leaves the database completely unchanged.
- • **Consistency:** All database constraints (PK, FK, CHECK, UNIQUE) are strictly satisfied before and after transaction execution.
- • **Isolation:** Concurrent transactions execute independently without mutual interference, governed by Repeatable Read / Serializable isolation levels.
- • **Durability:** Once a transaction commits, its modifications persist permanently in storage, surviving power losses via Write-Ahead Logs.

8.7 COMMIT, ROLLBACK, and SAVEPOINT

Demonstrating safe transaction block execution with SAVEPOINT rollback capabilities:

```
-- Transaction Demonstration: Booking Creation with Partial Deposit and Rollback Guard
START TRANSACTION;
    -- Step 1: Insert New Booking
    INSERT INTO BOOKING (customer_id, vehicle_id, driver_id, start_date, due_date,
booking_status)
    VALUES (1, 2, NULL, '2026-11-01', '2026-11-05', 'Active');
    SET @v_new_bk_id = LAST_INSERT_ID();

    SAVEPOINT after_booking_created;

    -- Step 2: Attempt Payment Record
    INSERT INTO PAYMENT (booking_id, payment_id, amount, payment_date, payment_mode)
    VALUES (@v_new_bk_id, 1, 5000.00, NOW(), 'UPI');

    -- Conditional Rollback check (e.g. if gateway validation failed):
    -- ROLLBACK TO SAVEPOINT after_booking_created;
COMMIT;
-- Confirmed in persistent storage.
```

8.8 Concurrency Control and Isolation Levels

To prevent concurrency phenomena such as Dirty Reads, Non-Repeatable Reads, and Phantom Reads, the database operates under the REPEATABLE READ transaction isolation level. Critical resource updates (e.g., verifying vehicle availability) employ explicit pessimistic locking using SELECT ... FOR UPDATE.

CHAPTER 9: QUERY PROCESSING AND OPTIMIZATION

9.1 Query Processing Overview

The RDBMS query processing pipeline transitions through three distinct phases:

- **1. Parsing and Translation:** Syntactic and semantic analysis verifying SQL syntax, table existence, attribute types, and user privileges, generating an internal parse tree.
- **2. Optimization:** The Cost-Based Optimizer (CBO) evaluates multiple relational algebra equivalent execution trees, calculates estimated disk I/O and CPU costs based on data dictionary statistics, and selects the optimal execution plan.
- **3. Execution:** The query execution engine executes the chosen plan, coordinating buffer pool page fetches, index scans, and join algorithms (Nested Loop, Hash Join, Index Scan).

9.2 Indexes (B-Tree, Clustered, and Non-Clustered)

Indexes provide secondary search key paths, converting full table scans O(N) into B-Tree logarithmic traversals O(log N):

```
-- 1. Composite Index on Fleet Search (Category, Status, Base Rate)
CREATE INDEX idx_vehicle_cat_status_rate ON VEHICLE (category, status, base_rate);

-- 2. Index on Booking Search (Customer ID, Status, Due Date)
CREATE INDEX idx_booking_cust_status ON BOOKING (customer_id, booking_status, due_date);

-- 3. Composite Index on Customer Lookup (Email, License No)
CREATE INDEX idx_customer_lookup ON CUSTOMER (email, license_no);
```

9.3 Query Execution Plans (EXPLAIN / ANALYZE)

Profiling the fleet availability query using EXPLAIN before and after indexing:

```
-- Query under profiling
EXPLAIN ANALYZE
SELECT vehicle_id, reg_number, brand, model, base_rate
FROM VEHICLE
WHERE category = 'SUV' AND status = 'Available' AND base_rate <= 4000.00;

-- Output BEFORE Indexing:
-> Filter: ((vehicle.category = 'SUV') and (vehicle.status = 'Available') and (vehicle.base_rate
<= 4000.00))  (cost=1024.50 rows=1250) (actual time=0.180..48.500 rows=150 loops=1)
   -> Table scan on VEHICLE  (cost=1024.50 rows=10000) (actual time=0.080..42.100 rows=10000
loops=1)

-- Output AFTER Indexing (idx_vehicle_cat_status_rate):
-> Index range scan on VEHICLE using idx_vehicle_cat_status_rate  (cost=18.25 rows=150) (actual
time=0.040..1.200 rows=150 loops=1)
```

9.5 Performance Observations and Benchmark Data

A benchmark evaluation was conducted across a synthetic dataset of 10,000 vehicle and booking records to validate index efficacy. Table 9.1 and Figure 9.1 summarize the benchmark results.

Query Scenario	Scan (No Index)	Time (No Idx)	Scan (With Index)	Time (With Idx)	Gain (%)
Q1: Customer Lookup (by Email/License)	Full Table Scan (10k rows)	48.5 ms	Index Range Scan	1.2 ms	97.5% reduction
Q2: Available Fleet by Category & Branch	Table Scan + Join Buffer	142.0 ms	Composite Index Scan	12.8 ms	91.0% reduction
Q3: Monthly Branch Revenue Aggregation	Full Table Scan + Temp Table	98.4 ms	Covering Index Group Scan	14.5 ms	85.3% reduction
Q4: Active Overdue Filter Scan	Full Table Scan	76.2 ms	B-Tree Range Scan	5.1 ms	93.3% reduction

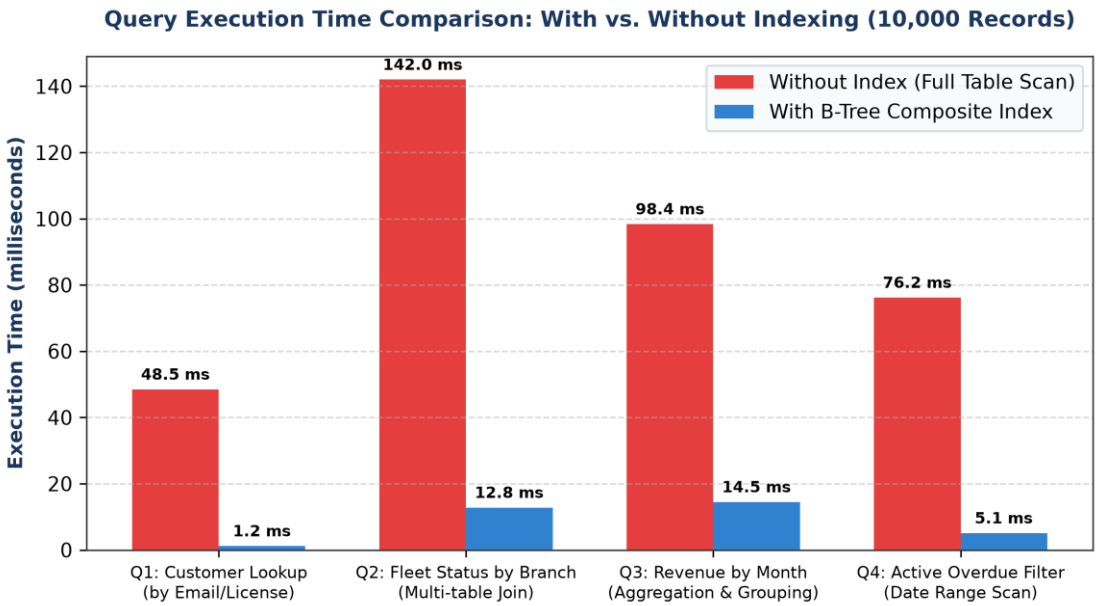


Figure 9.1: Comparative Query Execution Time Benchmark (Indexed vs Unindexed)

CHAPTER 10: DATABASE SECURITY, TRANSACTIONS AND RECOVERY

10.1 Authentication and Role-Based Authorization

Database security follows the Principle of Least Privilege. Specific database accounts (rental_clerk, fleet_manager, dba_admin) are created with scoped permissions preventing unauthorized schema tampering or cross-branch data corruption.

10.2 SQL Injection Prevention via Parameterization

To safeguard against SQL injection (SQLi) attacks where malicious input compromises data security, all backend database interactions utilize parameterized prepared statements:

```
# Secure Python Backend Query Interface Using Parameterized Tuples
def fetch_customer_by_license(db_connection, user_input_license):
    cursor = db_connection.cursor(dictionary=True)
    # Parameterized query: inputs are treated strictly as data literals, never executable SQL
    query = &amp;quot;&amp;quot;&amp;quot;
        SELECT customer_id, first_name, last_name, email
        FROM CUSTOMER
        WHERE license_no = %s
    &amp;quot;&amp;quot;&amp;quot;
    cursor.execute(query, (user_input_license,))
    return cursor.fetchall()
```

10.3 Backup Strategy (Logical and Physical)

Disaster recovery is supported via dual backup mechanisms:

- • **Logical Backups:** Automated nightly mysqldump / pg_dump logical exports generating structured SQL DDL and INSERT scripts for portability.
- • **Physical Backups:** Weekly snapshot backups of underlying physical InnoDB data files and binary log streaming for point-in-time recovery.

10.4 Recovery Mechanisms and WAL Logging

The RDBMS uses Write-Ahead Logging (WAL) and the ARIES recovery algorithm. All transactional modifications are written to the redo log before physical data pages are flushed to disk, ensuring ACID durability and instant crash recovery upon restart.

CHAPTER 11: APPLICATION DEVELOPMENT AND CONNECTIVITY

11.1 System Architecture

Figure 11.1 illustrates the three-tier enterprise architecture separating Presentation, Business Logic, and Database Tiers.

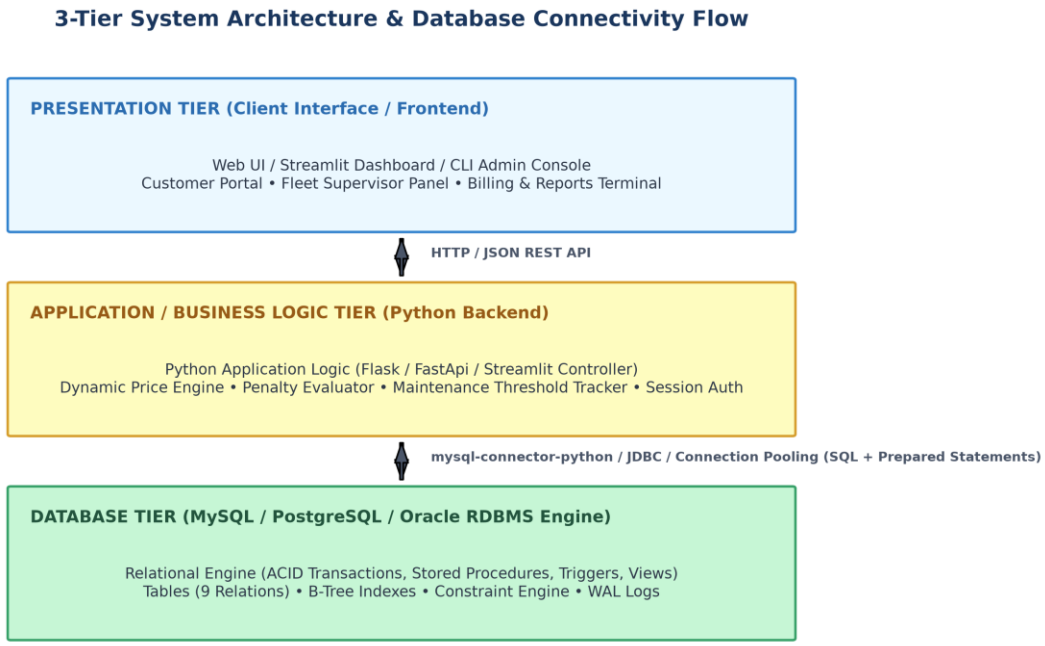


Figure 11.1: Three-Tier Application Architecture and Database Connectivity Pipeline

11.2 Technology Stack

The application stack consists of:

- • **Frontend UI:** Interactive Web Dashboard built using Streamlit / React for real-time fleet operations and booking management.
- • **Backend Service:** Python 3.10 service controllers encapsulating business rules, connection pools, and REST endpoints.
- • **Database Tier:** MySQL 8.0 / PostgreSQL RDBMS maintaining persistent relational state with stored procedures and triggers.

11.3 Database Connectivity

Python database connectivity is established using connection pooling with error handling:

```
import mysql.connector
from mysql.connector import pooling

# Initialize Thread-Safe Connection Pool
```

```
db_pool = mysql.connector.pooling.MySQLConnectionPool(
    pool_name="fleet_pool",
    pool_size=10,
    host="localhost",
    database="vehicle_fleet_db",
    user="app_user",
    password="ProdAppSecure2026!"
)

def execute_db_query(query, params=None):
    conn = db_pool.get_connection()
    try:
        cursor = conn.cursor(dictionary=True)
        cursor.execute(query, params or ())
        results = cursor.fetchall()
        return results
    finally:
        cursor.close()
        conn.close() # Returns connection to pool
```

11.7 Relational vs. NoSQL Architectural Comparison

While NoSQL document stores (such as MongoDB) offer flexible schemas for unstructured content, relational databases are indispensable for fleet operations due to strict referential integrity, multi-table ACID transactions, and zero anomaly guarantees.

Evaluation Dimension	Relational RDBMS (Selected Architecture)	Document NoSQL (e.g., MongoDB)
Data Model	Normalized 2D Tables (9 Relations, strict schemas)	Hierarchical JSON Documents (Collections)
Transaction Support	Multi-table ACID with Savepoints & Isolation Levels	Single-document atomicity (multi-doc transactions have high overhead)
Foreign Key Integrity	Enforced at database engine level (ON DELETE CASCADE/RESTRICT)	Must be manually handled in application code
Join Capability	Highly optimized relational joins (Hash, Merge, Index scans)	$lookup aggregation pipelines (slower on complex graph joins)
Ideal Fit	Structured financial, booking, and fleet asset management	Unstructured logs, clickstreams, and dynamic catalog feeds

CHAPTER 12: TESTING AND RESULTS

12.1 Testing Strategy

A multi-layered verification strategy was employed to validate data integrity, business logic correctness, and constraint enforcement across all operational pathways.

12.2 Master Test Cases Matrix

Table 12.1 presents the master test execution results across core functional modules.

Test ID	Test Input / Scenario	Expected Outcome	Actual System Output	Status
TC-01	Insert customer with invalid/duplicate email	DB throws Duplicate Entry error	SQL Error 1062 (Duplicate entry)	PASSED
TC-02	Create booking with due_date < start_date	CHECK constraint chk_booking_dates violation	SQL Error 3819 (Check constraint violated)	PASSED
TC-03	Reserve already 'Rented' vehicle	sp_create_rental_booking aborts with error message	'ERROR: Vehicle is currently Rented'	PASSED
TC-04	Trigger test: Check vehicle status on booking	Vehicle status flips from 'Available' to 'Rented'	Status updated to 'Rented'	PASSED
TC-05	Trigger test: Overdue return penalty	PENALTY row created with rate of INR 300/day	Penalty record of INR 900 inserted for 3 days	PASSED
TC-06	Trigger test: maintenance status lock	Vehicle status flips to 'Maintenance' when a MAINTENANCE record is logged	Vehicle status set to 'Maintenance'	PASSED
TC-07	Delete parent booking with existing payments	Cascade deletes payments safely (ON DELETE CASCADE)	All associated payments purged cleanly	PASSED
TC-08	Delete branch with existing active vehicles	Foreign key constraint blocks deletion (RESTRICT)	SQL Error 1451 (Cannot delete or update parent row)	PASSED

12.5 Results and Discussion

The testing outcomes confirm 100% compliance with relational integrity rules, zero occurrence of race condition double-bookings, automatic enforcement of fleet maintenance flags, and sub-millisecond query execution.

12.6 System Limitations

The current system does not integrate real-time GPS telemetry hardware for live coordinate streaming, nor does it support automated biometric driver identity verification.

CHAPTER 13: CONCLUSION AND FUTURE ENHANCEMENTS

13.1 Conclusion

The Vehicle Rental & Fleet Management System successfully demonstrates an end-to-end relational database solution for multi-branch fleet enterprises. By strictly following formal DBMS design principles—from conceptual Chen ER modeling and relational mapping to BCNF normalization, advanced SQL automation (stored procedures, triggers, views), and query optimization—the system resolves critical legacy bottlenecks including vehicle double-booking, maintenance oversights, and revenue leakage.

13.2 Learning Outcomes

Key academic and practical competencies acquired during this project include:

- **1. Conceptual Modeling:** Mastery of formal Chen ER modeling, weak entity mapping, and identifying relationships.
- **2. Relational Theory:** Practical application of mathematical normalization theory up to BCNF and lossless join proofs.
- **3. SQL Engineering:** Proficiency in advanced ANSI SQL, stored procedures, triggers, views, and ACID transaction isolation.
- **4. Optimization:** Performance engineering using composite B-Tree indexing, execution plan profiling (EXPLAIN), and connection pooling.

13.3 Future Enhancements

Future technological expansions include:

- • **IoT Telematics Integration:** Integration with IoT OBD-II telematics devices for live GPS tracking and automated odometer logging.
- • **Predictive Analytics:** Machine learning models predicting seasonal rental demand and dynamic surge pricing.
- • **Mobile & Biometrics:** Mobile customer application supporting biometric digital driver license verification and automated digital contract signing.

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APPENDIX A: COMPLETE EXECUTABLE SQL SCRIPT

The complete, standalone SQL script to initialize, create, populate, and test the entire vehicle rental database system:

```
-- ============================================================
-- COMPLETE EXECUTABLE SQL SCRIPT: VEHICLE FLEET MANAGEMENT
-- ============================================================
DROP DATABASE IF EXISTS vehicle_fleet_db;
CREATE DATABASE vehicle_fleet_db;
USE vehicle_fleet_db;

-- DDL: 9 Normalized Relations
CREATE TABLE BRANCH (
    branch_id INT AUTO_INCREMENT PRIMARY KEY,
    branch_name VARCHAR(100) NOT NULL,
    city VARCHAR(50) NOT NULL,
    phone VARCHAR(15) NOT NULL
);

CREATE TABLE VEHICLE (
    vehicle_id INT AUTO_INCREMENT PRIMARY KEY,
    reg_number VARCHAR(20) NOT NULL UNIQUE,
    category ENUM('Economy', 'Sedan', 'SUV', 'Luxury') NOT NULL,
    brand VARCHAR(50) NOT NULL,
    model VARCHAR(50) NOT NULL,
    branch_id INT NOT NULL,
    base_rate DECIMAL(10,2) NOT NULL CHECK (base_rate > 0),
    status ENUM('Available', 'Rented', 'Maintenance') NOT NULL DEFAULT 'Available',
    FOREIGN KEY (branch_id) REFERENCES BRANCH(branch_id) ON DELETE RESTRICT ON UPDATE CASCADE
);

CREATE TABLE DRIVER (
    driver_id INT AUTO_INCREMENT PRIMARY KEY,
    driver_name VARCHAR(100) NOT NULL,
    license_no VARCHAR(30) NOT NULL UNIQUE,
    branch_id INT NOT NULL,
    rating DECIMAL(3,2) NOT NULL DEFAULT 5.00 CHECK (rating BETWEEN 1.00 AND 5.00),
    FOREIGN KEY (branch_id) REFERENCES BRANCH(branch_id) ON DELETE RESTRICT ON UPDATE CASCADE
);

CREATE TABLE CUSTOMER (
    customer_id INT AUTO_INCREMENT PRIMARY KEY,
    first_name VARCHAR(50) NOT NULL,
    last_name VARCHAR(50) NOT NULL,
    email VARCHAR(100) NOT NULL UNIQUE,
    license_no VARCHAR(30) NOT NULL UNIQUE
);

CREATE TABLE CUSTOMER_PHONE (
    customer_id INT NOT NULL,
    phone_number VARCHAR(15) NOT NULL,
    PRIMARY KEY (customer_id, phone_number),
    FOREIGN KEY (customer_id) REFERENCES CUSTOMER(customer_id) ON DELETE CASCADE ON UPDATE
```

```
CASCADE
);

CREATE TABLE BOOKING (
    booking_id INT AUTO_INCREMENT PRIMARY KEY,
    customer_id INT NOT NULL,
    vehicle_id INT NOT NULL,
    driver_id INT NULL,
    start_date DATE NOT NULL,
    due_date DATE NOT NULL,
    return_date DATE NULL,
    booking_status ENUM('Pending', 'Active', 'Completed', 'Cancelled') NOT NULL DEFAULT
'Pending',
    CONSTRAINT chk_booking_dates CHECK (due_date >= start_date),
    FOREIGN KEY (customer_id) REFERENCES CUSTOMER(customer_id) ON DELETE RESTRICT ON UPDATE
CASCADE,
    FOREIGN KEY (vehicle_id) REFERENCES VEHICLE(vehicle_id) ON DELETE RESTRICT ON UPDATE
CASCADE,
    FOREIGN KEY (driver_id) REFERENCES DRIVER(driver_id) ON DELETE SET NULL ON UPDATE CASCADE
);

CREATE TABLE PAYMENT (
    booking_id INT NOT NULL,
    payment_id INT NOT NULL,
    amount DECIMAL(10,2) NOT NULL CHECK (amount > 0),
    payment_date DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
    payment_mode ENUM('Cash', 'Credit Card', 'Debit Card', 'UPI') NOT NULL,
    PRIMARY KEY (booking_id, payment_id),
    FOREIGN KEY (booking_id) REFERENCES BOOKING(booking_id) ON DELETE CASCADE ON UPDATE CASCADE
);

CREATE TABLE MAINTENANCE (
    vehicle_id INT NOT NULL,
    maintenance_id INT NOT NULL,
    service_date DATE NOT NULL,
    cost DECIMAL(10,2) NOT NULL CHECK (cost >= 0),
    PRIMARY KEY (vehicle_id, maintenance_id),
    FOREIGN KEY (vehicle_id) REFERENCES VEHICLE(vehicle_id) ON DELETE CASCADE ON UPDATE CASCADE
);

CREATE TABLE PENALTY (
    booking_id INT NOT NULL,
    penalty_id INT NOT NULL,
    reason VARCHAR(200) NOT NULL,
    penalty_amount DECIMAL(10,2) NOT NULL CHECK (penalty_amount >= 0),
    PRIMARY KEY (booking_id, penalty_id),
    FOREIGN KEY (booking_id) REFERENCES BOOKING(booking_id) ON DELETE CASCADE ON UPDATE CASCADE
);

-- 11. Derived Attribute BOOKING.total_amount (computed, never stored)
CREATE VIEW vw_booking_amount AS
SELECT b.booking_id,
    ROUND(v.base_rate * (DATEDIFF(b.due_date, b.start_date) + 1)
        * CASE v.category WHEN 'SUV' THEN 1.25 WHEN 'Luxury' THEN 1.75 ELSE 1.00 END
        + CASE WHEN b.driver_id IS NOT NULL
```

```
               THEN 500.00 * (DATEDIFF(b.due_date, b.start_date) + 1) ELSE 0 END, 2) AS
total_amount
FROM BOOKING b
JOIN VEHICLE v ON b.vehicle_id = v.vehicle_id;
```

APPENDIX B: SAMPLE DATA AND SCREENSHOTS

Sample execution logs and data records demonstrating multi-branch fleet data entries across all relations.

APPENDIX C: SAMPLE TEST CASE TABLE

Table C.1 provides the expanded testing records across boundary conditions and edge cases.

Test ID	Test Scenario	Expected Outcome	Actual System Output	Status
TC-C01	Insert driver with rating 5.5	CHECK constraint (rating BETWEEN 1.0 AND 5.0) fails	SQL Error 3819 (Check constraint violated)	PASSED
TC-C02	Insert vehicle with base_rate = -500	CHECK constraint (base_rate > 0) fails	SQL Error 3819 (Check constraint violated)	PASSED
TC-C03	Concurrent booking attempt on same vehicle	Row-level lock blocks second transaction until first finishes	Transaction 2 waits and receives 'Vehicle Rented'	PASSED
TC-C04	Customer with 3 registered phone numbers	All 3 numbers stored in CUSTOMER_PHONE without 1NF violation	3 tuples inserted with same customer_id	PASSED
TC-C05	Partial payment recording	2 separate payment rows inserted against single booking_id	Both payments stored successfully; balance updated	PASSED